

Pseudocode for Programs:-

1. Sequential Search (Linear Search)

START

Read array A and key

For $i = 0$ to $n-1$

If $A[i] == \text{key}$ then

print "Element found at position", $i+1$

stop

Print "Element not found"

END.

2. Sequential Search with comparison count

start

Read array, key

count = 0

for $i = 0$ to $n-1$

count++

If $A[i] == \text{key}$

print position, count

stop.

3) 2-D Matrix Search.

START

Read matrix, key

for each row

for each column

If $\text{matrix}[i][j] == \text{key}$

print row, column

stop

END.

4) All Occurrences Search.

```
START
Read array, key
count = 0
for i = 0 to n-1
If A[i] = key
PRINT position.
count ++
PRINT count
END.
```

5) First Occurrence Search

```
START
Read array, key
for i = 0 to n-1
If A[i] == key
PRINT i+1
STOP
END.
```

6) String Sequential Search.

```
START
Read names, key
for each name
If name == key
PRINT position
STOP
END.
```


⑦ Register Number Search:

```
START
Read Register array, key
for i = 0 to n-1
    If reg[i] == key
        PRINT position
STOP
END.
```

⑧ Brute force String Matching

```
START
Read text, pattern
for i = 0 to n-m
    j = 0
    While j < m AND match
        j++
    If j == m
        PRINT position
END.
```

⑨ Find All Pattern Occurrences.

```
START
Read text, pattern
For each shift
    Compare pattern
    If match
        PRINT position
END.
```


(10) Case - Inensitive Matching

```
START
count = 0
for each shift
  compare characters
  count++
  print count
END.
```

(11) Pattern Matching Comparison Count

```
START
count = 0
for each shift
  compare characters
  count++
  print count
END.
```

(12) Alignment Display :-

```
START
for each shift
  Display pattern position
  compare match/mismatch.
END.
```

(13) Comparison table

```
START
for each shift
  count comparisons
  Display shift, count, result
END.
```


⑭ Successful / Unsuccessful full Search

START
search pattern
IF found
print success
ELSE
print failure
END.

⑮ Complexity Analysis

START
Best case = $O(n)$
Average case = $O(n*m)$
Worst case = $O(n*m)$
space = $O(1)$
END.

⑯ Sentinel Search

START
Add key as sentinel
Compare case match
If before end.

⑰

Normal Search

START
perform normal search
count comparisons
Display results
END.

⑮ Search performance

START

Search all keys

count

display results

Analyze complexity

END.



⑯ Pattern Shifting

Start

shift pattern

compare with text

Record result

END

⑰ Complete Brute force

Start

Read text

For every shift

compare character

print comparison

print position

END

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